

ORIGINAL ARTICLE

The Motivational Impact of GenAI Tools in Language Learning: a Quasi-Experiment Study

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关键词: 自我决定理论 | 结构方程模型 | 人工智能聊天机器人 | 生成式人工智能工具 | 语言学习动机

ABSTRACT

The evolution of generative artificial intelligence (GenAI) technologies may have markedly influenced language learning. Through offering personalised support, GenAI-driven chatbots have potentially transformed how learners engage with and sustain their motivation in learning languages. Drawing on self-determination theory (SDT), this study explores the impact of using GenAI tools on the autonomous motivation of a group of Chinese university students learning English by examining how GenAI tool integration influences their basic psychological needs (BPN) of autonomy, competence and relatedness, and the development of their language motivation over a four-month academic term. A quasi-experimental design was employed for two groups ($N = 355$), with the experimental group incorporated GenAI tools into their learning routine, while the control group followed traditional classroom instruction. The structural model indicated a notable enhancement in the influence of Time 2 autonomy on Time 2 autonomous motivation, observed exclusively within the experimental group. The repeated measures ANOVA indicated that integrating GenAI tools significantly enhanced students' autonomous motivation compared to traditional methods. The findings suggest that strategic implementation of GenAI tools can effectively boost motivation in language learning, offering valuable insights for the consideration of their use in the educational system.

生成式人工智能(GenAI)技术的快速发展正对语言学习产生愈加深刻的影响,尤其是在语言学习的参与方式和动机维持方面。本研究基于自我决定理论,探讨GenAI工具的使用对中国大学生英语学习自主动机的影响,重点分析其对自主性、胜任感和关系感等基本心理需求的满足,以及在四个月学期内语言学习动机的发展变化。研究采用准实验设计,对355名学生进行对比分析,其中实验组将GenAI工具融入日常学习,而对照组则采用传统课堂教学模式。结构方程模型结果显示,在实验组中,时间点2的自主性对时间点2的自主学习动机产生了显著的正向影响。重复测量方差分析进一步表明,与传统教学方法相比,整合GenAI工具显著增强了学生的自主学习动机。本研究结果表明,GenAI工具的合理使用能够有效提升语言学习动机,为GenAI技术的教学应用提供了重要的实证支持。

1 | Introduction

The rapid advancement of generative artificial intelligence (GenAI) technology has attached dynamism and fluidity to the language learning landscape (Baskara 2023). By creating natural, human-like interactions with users, GenAI tools such as chatbots open up new ways of constituting, accessing, and interpreting language learning (Barrot 2023; Jeon et al. 2023; Kohnke et al. 2023), which, as a result, have become more multimodal and more mediated than ever before. They represent a pedagogical approach that harnesses the educational applications of cutting-edge technology and their impact on language learners' motivation. The effectiveness of such an approach lies in its affordance of responding contextually to a wide range of inquiries and prompts whilst offering personalised learning support and teaching services. Yet, the success of GenAI tools in language learning hinges on maintaining a balanced approach, where the technology augments rather than replaces human interaction. Overreliance on GenAI tools without due attention to pedagogical methods or the role of instructors may inadvertently relegate both instructors and language learners to passive roles in the language learning process, posing challenges for the holistic development of language learners.

A main reason for the increasing interest in the integration of GenAI tools in language learning is perhaps related to their effects on language learning motivation. Motivation stands as a cornerstone in language learning, influencing the dedication, persistence, and ultimate success of language learners (Dörnyei et al. 2015; Mercer and Dörnyei 2020). Understanding the dynamics of language motivation is thus of considerable importance, as it not only informs pedagogical practices but also unveils the complex web of factors shaping an individual's path toward linguistic proficiency. Where the context of educational technology is concerned, the analysis of technology affordances that may engage interest and learning has received particular attention within the field of language acquisition (Corno and Mandinach 2004). Though it may seem patently obvious and deeply ingrained that GenAI-assisted language learning would benefit language learners and boost their motivation, very little empirical research has been conducted to test these assumptions (Alamer and Al Khateeb 2023).

Given the circumstances, the purpose of this study is to examine the extent to which the use of GenAI tools may influence learners' language motivation, particularly in the context of autonomous motivation. The autonomous motivation is investigated from Deci and Ryan's (1985) self-determination theory (SDT), which posits that individuals are driven by their internal desires and values rather than external pressures. A further aim is to explore the temporal changes of the learners' language motivation under this GenAI technique over an extended period. Much past research has focused on how learners' environments impact their language motivation using cross-sectional methodologies, yet there is a notable lack of studies investigating the prolonged impacts of technology-assisted language learning over extended periods (Alamer and Al Khateeb 2023), such as over the course of an entire language module. This extended timeframe is crucial for a more comprehensive assessment of causal relationships and for understanding how GenAI tools can continue to impact students' motivation over an extended period, which provides a better

recognition of GenAI tools' ability to enhance and sustain learner motivation. If validated, it sheds light on how this technology can navigate the inherent difficulties and successes in language learning.

2 | Literature Review

2.1 | Self-Determination Theory in Language Learning

Self-determination theory (SDT) (Deci and Ryan 1985; Ryan and Deci 2000) is a comprehensive framework encompassing motivation and personality, centring on the degree of autonomy or external control evident in human behaviour: whether it is self-determined or imposed by external factors. It posits that individuals inherently possess a proclivity to seek challenges and assimilate new experiences into their self-concept. The social environment therefore plays a crucial role, either fostering self-determined actions or hindering one's innate inclination to govern one's psychological development. Autonomy is manifested when individuals align their actions with their own volition. However, external influences, such as rewards and prescribed objectives, exert a controlling impact on individual behaviour. The essence of self-determination, as outlined by Deci (1980, 5), lies in the operation of will—"the capacity to choose behaviours based on inner desires and perceptions." Within the framework, there exist two broad categories of motivated behaviour: (1) self-determined and selected, guided by internalised motives and decisions, and (2) behaviours that are not chosen but seem to be influenced by conditioning, driven by controlling factors. In a sense, SDT places significant emphasis on a continuum of self-determination, directing focus on the mechanisms by which motivation for goals and behaviours defined by external factors can become socialised and progressively internalised (Dörnyei and Ushioda 2021). To foster and sustain autonomous motivation, it is essential to satisfy three basic psychological needs (BPN): autonomy, competence, and relatedness.

It was not until the early 2000s that SDT was applied to language learning and spawned a wealth of relevant research. The significant development of SDT in the realm of language learning can be attributed largely to the contributions of Kim Noels (Noels 2001, 2003; Noels et al. 1999, 2000). In particular, Noels et al. (2000) took the initiative to create a novel, second language (L2)-specific tool for evaluating the orientations of L2 learners through the lens of SDT. They further revealed that the identified regulation scale, which reflects goals and motives that are personally valued and internalised, showed a stronger relationship with the criterion variables compared to the intrinsic motivation subscales. According to Noels et al. (2000), factors like enjoyment and interest alone may not be enough to support sustained learning. Instead, the personal importance and value placed on learning the language might play a more crucial role in this context. Accordingly, the autonomous motivation can be well interpreted from two sub-constructs: intrinsic orientation and identified orientation, each emphasising the learner's self-perception throughout the language learning process.

More recently, BPN satisfaction and frustration have been shown to mediate the relationship between external environmental factors and motivational or behavioural outcomes (Al-Hoorie et al.,

2022; Ryan and Deci 2017). In language learning, satisfaction of BPN promotes motivational regulation and engagement, whereas BPN frustration, such as feeling controlled, incompetent, or isolated, can undermine motivation and hinder learning (Jang et al. 2016). Recent research has provided empirical evidence that BPN satisfaction correlates with more adaptive motivational regulation and better well-being outcomes. (Al-Hoorie et al., 2022). For instance, when students' BPN are met, they experience higher levels of self-confidence and lower failure avoidance (Lou and Noels 2020), further demonstrating how BPN satisfaction fuels autonomous motivation. Noels (2023) expanded on this by introducing a conceptual model that highlights how BPN satisfaction drives both linguistic proficiency and broader outcomes like sociocultural integration and psychological well-being. This model emphasises the role of contextual factors—such as teacher support, family involvement, and the wider language community—in either nurturing or hindering motivation. The model suggests that language learning takes place in diverse sociocultural and sociopolitical contexts, where external support plays a crucial role in sustaining BPN satisfaction and, by extension, autonomous motivation. These complexities of motivational dynamics in language learning were supported by Alamer et al. (2023), who argued that unmet BPN, such as a lack of competence, relatedness, or autonomy, can lead to diminished intrinsic motivation, impeding learners' optimal engagement and performance. They introduced the basic psychological needs frustration in second language (BPNF-L2) scale, which demonstrates that frustration in autonomy, competence, and relatedness can undermine intrinsic motivation and second language achievement. When BPN are frustrated—whether through perceived control, incompetence, or isolation—the negative impacts on motivation are more severe. BPN frustration can lead to maladaptive behaviour, passivity, and decreased engagement, significantly hindering the language learning process (Reeve 2022). This distinction is critical in understanding how external factors, such as teacher behaviour or classroom dynamics, can either support or disrupt learners' intrinsic motivation.

2.2 | AI-Mediated Instruction and Language Learning Motivation

AI-mediated instruction refers to the use of artificial intelligence technologies, such as generative AI and natural language processing (NLP), to facilitate personalised learning experiences and engage students in meaningful interaction. These AI-driven tools go beyond traditional language learning applications by providing real-time feedback, adaptive content, and tailored support that mimics human-like communication (Jeon et al. 2023; Kohnke et al. 2023). Students interact with AI-mediated tools similarly to how they would with human instructors, benefiting from adaptive learning that responds to their proficiency level and learning style. (Forsyth et al. 2019; Ockey and Chukharev-Hudilainen 2021)

A hallmark of current thinking about AI-mediated instruction is absorption of the seeming omnipresence of GenAI tools at the moment. GenAI tools refer to substantial generative language models, which possess a distinctive capability highlighted by their aptitude for generating text based on provided input prompts (Barrot 2023; Debby et al., 2024). It excels in creating chatbots

capable of engaging in natural language conversations, answering questions, and showcasing a profound understanding of diverse inputs (Debby et al., 2024). Given this feature, GenAI tools hold significant potential in language learning and instruction by offering personalised and interactive learning experiences. Leveraging its extensive vocabulary and contextual response generation, GenAI tools facilitate the development of communicative proficiency in a foreign language.

Given their potential to transform language learning environments, discussions about GenAI tools in the context of language learning motivation have thus far been predominantly situated within AI-mediated informal digital learning of English (AI-IDLE) (Guan et al. 2024; Liu et al. 2024). For instance, He et al. (2023) argued that students' "endogenous motivation of learning" is enhanced by positive perceptions of GenAI-supported learning experiences in informal contexts. Lee and Lee (2021) surveyed 661 Korean L2 English learners and found that their ideal L2 and ought-to L2 selves correlated strongly with their participation in IDLE activities. Similarly, Liu et al. (2024) confirmed that a clear ideal L2 self-image and a sense of enjoyment facilitated learners' adoption of digital tools, such as wearable devices, to perform IDLE activities.

Despite these insights, there is a noticeable gap in the literature regarding the application of GenAI tools in blended environments that integrate both formal and informal learning contexts. Moreover, research on GenAI-mediated language learning has predominantly examined motivation as an outcome variable but has not sufficiently explored how autonomous motivation—the internal drive to engage in activities for their inherent satisfaction—shapes learners' engagement in GenAI-mediated language learning environments. This gap is significant given that autonomous motivation is central to sustained and meaningful engagement in learning activities, particularly in blended settings where learners navigate between formal and informal learning environments.

2.3 | Purpose of the Study

Against the backdrop mentioned above, this study explores the role of GenAI tools in the autonomous motivation of a cohort of Chinese university students learning English. These students are traditionally associated with "passive learners" in their early learning stages (Liu and Qi 2005; Tran 2013), and as concomitants of such cliché, it is sometimes questionable whether they can truly take autonomous learning actions. The second aim is to, with the aid of a path analysis technique, examine the potential evolution of student language motivation over time under the use of GenAI tools. Accordingly, two research questions are articulated as follows:

RQ1. Could incorporating GenAI tools in language learning lead to a markedly stronger influence of BPN on students' autonomous motivation?

RQ2. Does the integration of GenAI tools in language learning, as opposed to traditional classroom learning,

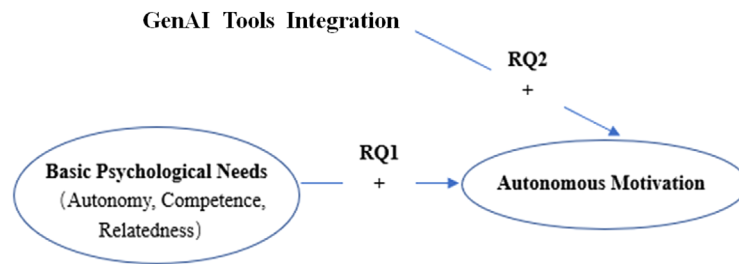


FIGURE 1 | The conceptual framework of the effect of GenAI tools on language learning motivation. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jlil.12701)]

result in an improvement in students' autonomous motivation?

To answer these questions, this study has employed Alamer and Al Khateeb's (2023) approach of examining the relationship between BPN and autonomous motivation over time. A conceptual framework is proposed in Figure 1.

3 | Method

This study employed a quasi-experimental design involving a control group and an experimental group at two different time intervals to examine how using GenAI tools in language learning affects students' language motivation and, specifically, to investigate whether the use of such a tool will encourage their BPN. We designed the intervention to involve GenAI tools as learning tools over the course of a four-month academic term, during which the control group did not undergo any intervention and remain in traditional learning conditions without using GenAI tools, whereas the experimental group was exposed to the intervention. It should be noted that the involvement of GenAI tools in language courses was designed and assessed by the instructors with due care, ensuring alignment with the content covered in the existing courses. Therefore, the utilisation can be seen as a change in the way of learning traditional materials to promote the goals of the courses and diversify the tasks assigned to students.

3.1 | Research Context and Participants

The study was conducted amongst the undergraduate students from two different campuses at a high-level research university in East China. These students are in the third year of the English major, who are mandated to enroll in the Advanced English course, which comprises two 100-min sessions per week, spanning approximately sixteen weeks throughout the autumn semester. The course is designed to train students in comprehensive English skills, particularly in reading comprehension, grammar, rhetoric, and writing.

355 students have participated in the study, with 141 assigned to the experimental group and 214 to the control group. The subset of the participants comprises all students enrolled in the Advanced English course across the two campuses during the research period. The allocation of the experimental and control groups were based on the natural composition of classes across campuses, with each group comprising four classes from the same

campus. Both groups consisted of students who had recently passed the Test for English Major 4 test, indicating a similar level of language proficiency. The average scores for both groups were very close, with no significant differences between them ($p = 0.56$). Regarding demographics, both groups were predominantly female (55% in each group) and had similar age distributions (ages 18-22). These factors were consistent across both groups, ensuring comparability on demographic variables. Additionally, we assessed language motivation levels of participants prior to the intervention using a SDT scale (See 3.2 and 3.3). The results indicated no significant differences between the groups in terms of overall motivation ($p = 0.455$). Specifically, the mean scores for autonomy ($M = 4.068$, $SD = 0.815$ for experimental; $M = 4.044$, $SD = 0.845$ for control), competence ($M = 4.152$, $SD = 0.894$ for experimental; $M = 4.169$, $SD = 0.831$ for control), and relatedness ($M = 3.673$, $SD = 0.924$ for experimental; $M = 3.756$, $SD = 0.860$ for control) were comparable, further suggesting that the groups were equivalent in terms of motivation prior to the intervention. A profile of the participants is shown in Table 1. Participants in the experimental group have access to GenAI tools on their personal digital devices.

3.2 | Instruments

The study consulted two scales from Alamer (2022) to collect the motivational information from the participants (see Appendix). The first scale was employed to assess the participants' BPN in three distinct dimensions: autonomy, competence, and relatedness. It comprises a total of 15 items and employs a seven-point Likert-type response format, ranging from one (strongly disagree) to 7 (strongly agree).

The second scale was adopted to measure the participants' autonomous motivation (Alamer 2022). The scale consists of ten items and was used to evaluate students' intrinsic orientation and identified orientation. The items were rated using a 7-point Likert scale ranging from one (strongly disagree) to seven (strongly agree).

3.3 | Procedures

After providing the consent form, participants were divided into either the experimental group or the control group depending on the campus where they were situated. They were then instructed to complete the initial questionnaire Time 1 (T1) during a class session in the first week of the semester.

TABLE 1 | Study participants (control and experimental groups).

Variable	Attribute	Follow-up (<i>n</i> = 286)		Attrition (<i>n</i> = 69)		Total sample		χ^2	<i>p</i>
		<i>n</i>	%	<i>n</i>	%	<i>n</i>	%		
Group	Control	118	41.3	23	33.3	141	39.7	1.458	0.227
	Experimental	168	58.7	46	66.7	214	60.3		
Gender	Male	126	44.1	34	49.3	160	45.1	0.612	0.434
	Female	160	55.9	35	50.7	195	54.9		

Note: Attrition refers to the participants who were initially part of the study but were unavailable for follow-up or dropped out of the study before its completion.

The primary responsibility of participants in the experimental group was to utilise GenAI tools to aid in their English course learning. To facilitate this process, an orientation meeting was organised for the students in the experimental group, during which the researchers offered comprehensive details about the intervention activities and provided instructions on how to effectively engage with GenAI tools throughout the course. Specifically, participants were instructed to follow a structured 4-step learning cycle for each unit: firstly, they were directed to develop their own learning plan with GenAI tools using the upcoming lesson topic as the prompt, brainstorming ideas and outlining the lesson plan accordingly. According to SDT, autonomy is a central principle in fostering intrinsic motivation. By allowing students to freely explore and generate ideas, this phase promotes a sense of volition and ownership over their learning. GenAI tools' conversational interface supports this process by providing an open-ended, low-pressure environment in which students can express their ideas without fear of judgment. Its ability to generate multiple suggestions and ask probing questions further encourages students to engage in self-directed thinking, empowering them to take charge of their ideation process. After that, students were required to critically review the output from GenAI tools and refine the learning plan based on their preferences and the specific objectives of the unit. In this phase, students used GenAI tools to refine their ideas, seeking clarifications or alternative perspectives. This phase aligns with competence, as it encourages students to reflect on their initial ideas and improve them through iterative feedback. GenAI tools serve as a collaborative partner in this process, offering suggestions and posing critical questions that challenge students' thinking. The ability to continuously refine their responses boosts students' confidence and fosters a sense of progress and mastery over the content. During class sessions, activities such as discussions were facilitated based on the learning plan outlines generated. The third phase encouraged students to share their refined ideas with peers using GenAI tools, either through group discussions or collaborative document editing. This phase is rooted in SDT's concept of relatedness, as students work together to build upon each other's ideas. GenAI tools facilitate this collaboration by offering suggestions for further exploration, fostering a sense of community, and encouraging peer learning. The final phase of the learning cycle involved students presenting their ideas in a final format, such as an essay or oral presentation, and receiving feedback. This phase connects to SDT's principles of competence and relatedness by allowing students to demonstrate their learning achievements and receive constructive feedback from both peers and instructors. GenAI tools' feedback capabilities, such as offering suggestions for improvement or posing questions to

provoke deeper analysis, encourage students to view the feedback process as a tool for further growth. This phase also promotes autonomy by giving students the choice in how they present their work, allowing for self-expression in the final product. This learning cycle was repeated six times throughout the semester to align with the teaching content (comprising six units). During each class session, experimental group students were instructed to interact with GenAI tools for approximately fifteen minutes, completing one step in each learning cycle. It should be noted that the key tenet underlying the above design is not merely the use of GenAI as a tool, but rather its implementation into our structured four-step learning cycle during class session.

It is also important to note that both experimental and control group students were taught by the same instructor and received identical instruction within the classroom environment, aside from utilising GenAI tools. During the process, the instructor was aware of the fact that while GenAI tools offer valuable assistance and support for language learning, at this stage, it cannot substitute human interaction and feedback necessary for effective language acquisition. Therefore, the instructor took GenAI tools as a supplement to students' learning rather than a replacement for it. At the end of the semester, in Week 16, all participants, encompassing both the experimental and control groups, were asked to complete the same questionnaire Time 2(T2).

3.4 | Data Analysis

The collected data were analysed using SPSS 27.0 and AMOS 26.0, following an array of analytical procedures to ensure rigorous statistical evaluation. Descriptive analyses were first performed: categorical variables were described using frequency and percentage, while continuous variables were summarised with mean and standard deviation. The distribution characteristics of the variables were presented using kurtosis and skewness (Kline 2016). To assess the quality of the scale data, Harman's single-factor test was employed to check for common method bias (Podsakoff et al. 2003), and Cronbach's alpha was used to measure reliability (Cronbach 1951). Validity was evaluated through confirmatory factor analysis (CFA) (Hair et al. 2018), with composite reliability (CR) and average variance extracted (AVE) used to represent convergent validity. As recommended by Kline (2016), the model's fit was assessed using the χ^2/df ratio, the root mean square error of approximation (RMSEA), the comparative fit index (CFI), and the Tucker-Lewis index (TLI). A good model fit is indicated by a χ^2/df

ratio of less than 5, an RMSEA of less than 0.08, and CFI and TLI values greater than 0.90 (Kline 2016). To examine the relationships between variables at different time points, correlation analyses were then conducted using Pearson correlation. Following that, structural equation modelling (SEM) was carried out using AMOS 26.0 to explore the mechanisms through which BPN influenced autonomous motivation, comparing the control group and the experimental group. Additionally, repeated measures ANOVA were employed to assess the intervention effects of different methods on autonomous learning motivation. The significance level was set at 0.05 ($p < 0.05$), indicating statistical significance of the observed differences.

4 | Results

4.1 | Outliers and Normality

Mahalanobis distance (D^2) was employed to identify multivariate outliers, measuring the distance between cases in terms of variance units (Kline 2016). Given that D^2 follows a chi-square distribution and this study includes 8 dependent variables, the critical value for $p = 0.001$ with 8 degrees of freedom is 26.13. A maximum D^2 value of 24.413 was found in this study, which is below the critical value. Therefore, the study does not exhibit multivariate outliers.

Skewness and kurtosis were considered to describe the normality of data distribution (Kline 2016). According to Hair et al. (2021), for a normal distribution, absolute values of skewness and kurtosis less than 1 are considered excellent, while absolute values less than 2 are deemed acceptable. In this study, the absolute values of skewness and kurtosis for all variables were less than 1, indicating that the data can be considered normally distributed.

4.2 | Reliability and Validity

The study drew on Harman's single-factor test to examine common method bias, including all items from the scales in a factor analysis. The variance contributions of the unrotated first factor at the pre-test (T1) and post-test (T2) time points were 34.885% and 32.005%, respectively, both below the 50% threshold (Podsakoff 2003), indicating that serious common method bias was not present in this study.

The results of the reliability and validity for the pre-test and post-test are shown in Table 2. The Cronbach's Alpha for each variable met the 0.7 standard, indicating good reliability and internal consistency of the variables at each time point in this study. A CFA model was established using AMOS 26.0. The analytical results showed that all fit indices for the pre-test model met the fit standards ($\chi^2 = 337.346$, $df = 199$, $\chi^2/df = 1.695$, $p < 0.001$, RMSEA = 0.049 [0.040, 0.058], CFI = 0.959, TLI = 0.953, SRMR = 0.049). Similarly, the post-test CFA model also met the fit standards ($\chi^2 = 369.877$, $df = 199$, $\chi^2/df = 1.859$, $p < 0.001$, RMSEA = 0.055 [0.046, 0.064], CFI = 0.945, TLI = 0.936, SRMR = 0.043). In addition, the CR for each variable was greater than 0.7, and the AVE was greater than 0.5, indicating good composite reliability and convergent validity for all variables in this study. The Pearson correlation

TABLE 2 | Validity and reliability of the construct.

Construct	CR	AVE	Cronbach's alpha
T1			
Autonomy	0.851	0.591	0.841
Competence	0.840	0.567	0.839
Relatedness	0.887	0.662	0.885
Autonomous motivation			0.897
Intrinsic orientation	0.880	0.597	0.879
Identified orientation	0.882	0.599	0.880
T2			
Autonomy	0.856	0.598	0.849
Competence	0.816	0.526	0.814
Relatedness	0.841	0.571	0.841
Autonomous motivation			0.896
Intrinsic orientation	0.876	0.587	0.874
Identified orientation	0.882	0.601	0.877

analysis shows good internal consistency for all variables over 16 weeks, with Cronbach's alpha coefficients greater than 0.7 at Time 1 and Time 2 across both groups.

We also conducted longitudinal measurement invariance testing for the constructs. The fit results for configural invariance, weak invariance, and strong invariance models are shown in Table 3. As is shown, the chi-square test between the configural invariance model and the weak invariance model was not significant ($\Delta\chi^2 = 21.893$, $\Delta df = 17$, $\Delta P = 0.189$, $\Delta TLI = -0.002$, $\Delta CFI = 0.001$), and there was no significant difference between the weak invariance model and the strong invariance model ($\Delta\chi^2 = 12.405$, $\Delta df = 15$, $\Delta P = 0.648$, $\Delta TLI = -0.002$, $\Delta CFI = -0.001$). Accordingly, the structure of the motivational constructs has good stability over Time 1 and Time 2.

4.3 | Structure Model Testing Between the Control and Experimental Groups

Drawing on two independent sample t -tests, no significant differences were found between the experimental groups at the pre-test (T1) time point in terms of autonomy ($t = -0.241$, $p = 0.810$), competence ($t = 0.170$, $p = 0.865$), relatedness ($t = 0.776$, $p = 0.438$), and autonomous motivation ($t = 0.502$, $p = 0.616$) ($p > 0.05$). This indicates that the baseline levels of the variables involved in this study were equivalent, meeting the experimental conditions.

RQ1. Could incorporating GenAI tools in language learning lead to a markedly stronger influence of BPN on students' autonomous motivation?

To answer this question, we used statistical models designed to compare different groups, called multigroup path models. These models allowed us to explore whether the relationships between BPN and autonomous motivation differed between

TABLE 3 | Longitudinal measurement invariance test.

Model	χ^2	df	<i>p</i>	χ^2/df	RMSEA	TLI	CFI
(a) Configural invariance	707.223	398	<0.001	1.777	0.037[0.032,0.041]	0.944	0.952
(b) Metric invariance	729.116	415	<0.001	1.757	0.036[0.032,0.041]	0.946	0.951
(c) Scalar invariance	741.521	430	<0.001	1.724	0.036[0.031,0.040]	0.948	0.952

Note: Configural invariance: Tests whether the configuration or pattern of the latent variables is the same. weak invariance (also known as metric invariance): Tests whether the factor loadings of the measurement indicators are equivalent across groups. strong invariance (also known as scalar invariance): Tests whether the intercepts of the observed variables are invariant.

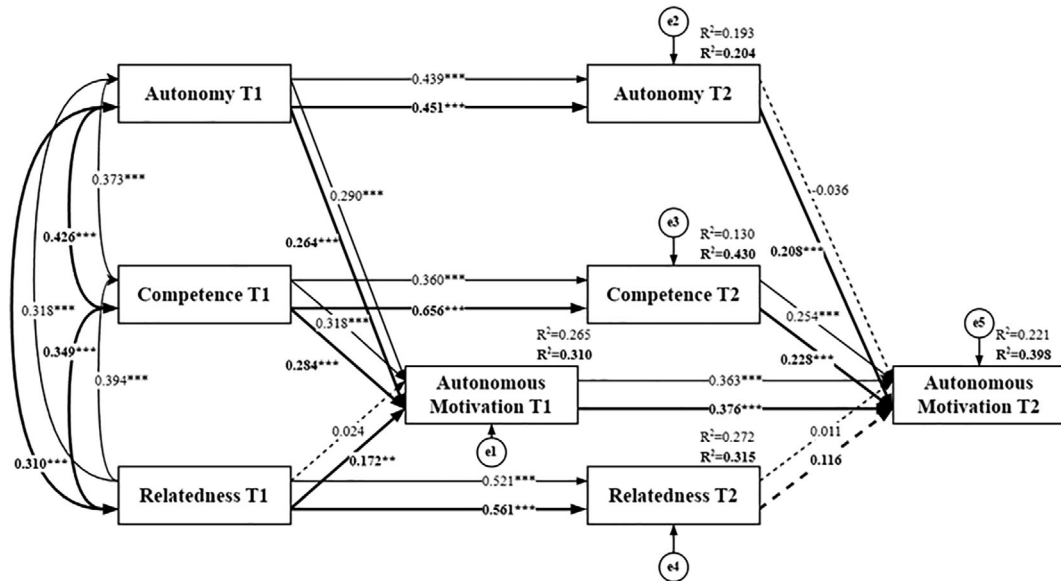


FIGURE 2 | The results of the path analysis in control and experimental groups. * indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.001$. The coefficients are standardised. Thin lines represent results for the control group, while thick lines represent results for the experimental group; Solid lines indicate significant effects, whereas dashed lines indicate non-significant effects.

the experimental and control groups. To ensure the comparison was meaningful, we first checked that the constructs we measured at two different times (T1 and T2) were consistent across both groups, a process known as testing for measurement invariance. The results of the path analysis are presented in Figure 2. Following previous studies (Guay et al. 2015; Tóth-Király et al. 2017), we first tested for overall model differences. If significant differences were found between the experimental and control group models, it would indicate that the treatment caused overall changes in the model, and we would further examine the differences in the model paths. Upon identifying significant path differences, we proceeded to compare the path coefficients.

Given the above procedures, an examination of the multigroup invariance tests for the overall model was undertaken. The results showed that the unconstrained model demonstrated a good fit ($\chi^2 = 46.591$, $df = 24$, $\chi^2/df = 1.941$, $p = 0.004$, RMSEA = 0.058 [0.032, 0.082], CFI = 0.966, TLI = 0.920), and the structurally constrained model also met the fit criteria ($\chi^2 = 66.663$, $df = 34$, $\chi^2/df = 1.961$, $p = 0.001$, RMSEA = 0.058 [0.037, 0.079], CFI = 0.950, TLI = 0.918). Comparing the models, there was a significant difference between the unconstrained model and the structurally constrained model ($\Delta\chi^2 = 20.071$, $\Delta df = 10$, $\Delta P = 0.029$, $\Delta TLI = 0.002$).

Subsequently, a multigroup invariance test was conducted for each structural path. Specifically, there was a notable difference in the influence of T1 competence on T2 competence ($\Delta\chi^2 = 9.293$, $\Delta df = 1$, $\Delta P = 0.002$), with the experimental group ($\beta = 0.656$, $p < 0.001$) showing a significantly stronger effect than the control group ($\beta = 0.360$, $p < 0.001$). In other words, students in the experimental group who had higher initial competence (T1) experienced a greater improvement in their competence over time (T2) compared to students in the control group. Additionally, there was a significant difference between the control and experimental groups in the impact of T2 autonomy on T2 autonomous motivation ($\Delta\chi^2 = 6.167$, $\Delta df = 1$, $\Delta P = 0.013$), where the experimental group's T2 autonomy had a significantly stronger effect on T2 autonomous motivation ($\beta = 0.208$, $p = 0.001$) compared to the control group ($\beta = -0.036$, $p = 0.670$). This indicates that in the experimental group, students' sense of autonomy at T2 significantly and positively influenced their autonomous motivation at this point, whereas in the control group, this relationship was not significant and was slightly negative. Meanwhile, the model's predictive power (R^2) indicated that the experimental group model accounted for roughly 39.8% of the variance in autonomous motivation ($R^2 = 0.398$), whereas the control group model explained about 22.1% ($R^2 = 0.221$). This indicates that the experimental model, which includes the

TABLE 4 | Results of the repeated measures ANOVA.

Source	Type III sum of squares	DF	Mean square	F	<i>p</i>	η_p^2
Tests of between-subjects effects						
Intercept	8086.724	1	8086.724	9806.856	<0.001	0.972
Treatment	3.763	1	3.763	4.563	0.034	0.016
Error	234.186	284	0.825			
Tests of within-subjects contrasts						
Time	0.699	1	0.699	2.531	0.113	0.009
Time $\times$ Treatment	6.112	1	6.112	22.142	<0.001	0.072
Error (time)	78.390	284	0.276			

Note: η_p^2 = Partial eta squared.

intervention of using GenAI tools, is more effective in explaining the variance in students' autonomous motivation.

RQ2. Does the integration of GenAI tools in language learning, as opposed to traditional classroom learning, result in an improvement in students' autonomous motivation?

To determine if incorporating GenAI tools with the experimental group led to a rise in autonomous motivation from T1 to T2, compared to the control group, a repeated measure ANOVA was utilised.

The results reported in Table 4 indicated a significant difference in autonomous motivation between the experimental group and the control group ($F(1, 284) = 4.563, p = 0.034, \eta_p^2 = 0.016$). While the observed effect size is small according to Cohen's (1988) guidelines ($\eta_p^2 = 0.02$), this still suggests that the intervention had a measurable impact on autonomous motivation. Moreover, there was a significant interaction effect between the interventions and the measurement time points ($F(1, 284) = 22.142, p < 0.001, \eta_p^2 = 0.072$). This medium effect size ($\eta_p^2 = 0.07$) indicates that the experimental group experienced a meaningful increase in autonomous motivation over time relative to the control group. The slope in Figure 3 depicts the variations in autonomous motivation over time for the two groups. This pattern aligns with the results presented in Table 4, which indicate a notable rise in autonomous motivation amongst the participants in the experimental group as a consequence of the intervention, whereas members of the control group experienced a decrease in autonomous motivation over time.

Further simple effects tests were conducted, and the results showed that at T1, there was no significant difference between the control group and the experimental group ($p = 0.616$). The autonomous motivation scores of the experimental group were significantly higher than those of the control group at T2 (mean difference = $-0.375, p < 0.001$). Additionally, tests for differences between pre-test and post-test scores in different groups revealed that the post-test autonomous motivation scores of the control group were significantly lower than their pre-test scores (mean difference = $0.139, p = 0.043$), while the post-test autonomous motivation scores of the intervention group were significantly

higher than their pre-test scores (mean difference = $-0.281, p < 0.001$). In summary, the results of the repeated measures ANOVA indicate that the intervention implemented in this study significantly enhances students' autonomous motivation.

5 | Discussion

The study was designed to assess how effectively GenAI tools could potentially enhance our understanding of students' motivation for language learning. To this end, we drew on structural modelling to investigate the relationship between students' BPN (autonomy, competence, and relatedness) and their autonomous motivation over time. We also examined whether such implementation could directly lead to an increase in autonomous motivation amongst the students who used GenAI tools over the course of the language learning compared to their counterparts.

Specifically, the structural model (Figure 2) demonstrated an increasing impact of autonomy on autonomous motivation over time within the experimental group. This indicates that integrating GenAI tools into language learning contributed to the observed rise in autonomous motivation in the experimental group, as opposed to the control group in the post-test. It is also revealed that the initial competence level more strongly affected subsequent competence in the experimental group, though its influence on autonomous motivation was not as pronounced compared to the other BPNs. Additionally, our results, verified through repeated measures ANOVA, confirmed a unique increase in autonomous motivation over time within the experimental group. The statistical analyses provide new insights into the directional relationship between GenAI usage in language learning and student motivation.

Despite the assertion that the complete set of BPN is essential for developing learners' autonomous motivation (Ryan and Deci 2020), it appears that, given the results of this study, GenAI tools can boost students' autonomous motivation by only enhancing their autonomy. This finding has been somehow supported by Alamer et al. (2023) who found that technology-assisted language learning applications can be seen as a language learning medium that promoted autonomous motivation by fostering a sense of autonomy. In this sense, students who incorporated GenAI tools in their language learning were convinced that

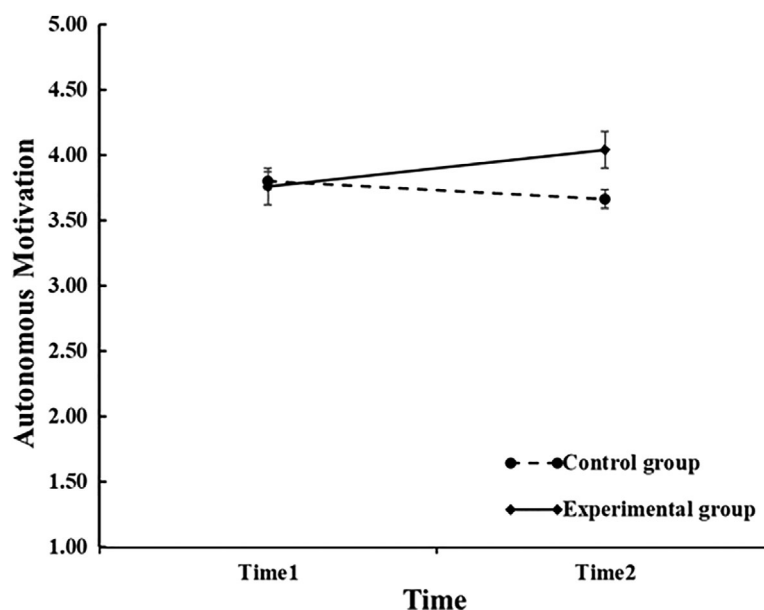


FIGURE 3 | Variations in students' autonomous motivation over time between the experimental and control groups.

they were engaged in the learning with genuine interest and commitment and were fully aware of their ability to take control of the learning process. This is partially because, as suggested by Kartal (2019), integrating interactive applications in language learning environments aids in alleviating learners' anxiety and therefore improving their autonomy and language skills. GenAI tools offer a non-judgmental environment where students can communicate without fear of making mistakes, which largely boosts students' confidence and willingness to take initiative in their learning. Another important reason is that GenAI tools may potentially encourage learners' curiosity and exploration. By asking open-ended questions and engaging in conversations, students are likely to explore deeper into topics of interest, which, in turn, cultivates a sense of ownership. Moreover, the affordances of GenAI chatbots allow for personalised learning experiences. Students can access a wide range of language learning resources and interact with GenAI tools at their own pace, allowing them to ask questions and receive instant feedback. This self-directed learning approach and the ease of access support autonomy by assigning students the freedom to decide when, what, and how they learn (Crompton and Burke 2024). It is particularly evident in our research context, where students in the experimental group are required to use GenAI tools to complete tasks including four dimensions of autonomous learning: self-direction, self-pacing, self-assessment, and choice and control (see Section 3.3) during their in-class language learning.

Similar findings were not observed for the other two BPNs. There were minimal differences in the effects of competence and relatedness on autonomous motivation between the two groups in the post-test. This is consistent with the empirical results of Alamer and Al Khateeb (2023) but contradicts the findings of Karapanos et al. (2016), who asserted that competence and relatedness are two of the most crucial needs within a technology-enhanced language learning environment. In terms of learners' competence while using GenAI tools, interestingly, it was found that students in the experimental group who had higher initial competence (T1)

tended to experience a greater improvement in their competence over time (T2). This can be explained by GenAI tools' ability to provide language learners with personalised learning plans and appropriately challenging tasks. For students who start with a higher level of competence, GenAI tools offer more advanced challenges, helping them to further improve and feel even more capable. As a result, regular successful interactions reinforce students' belief in their abilities, especially for those with higher initial competence who experience frequent successes. However, it seems that this improvement in competence did not directly translate into a notable increase in the autonomous motivation. A possible explanation is that competence involves not only recognising one's success but also feeling appreciated and validated in a social context (Ibarra-Sáiz et al. 2020). While GenAI tools help improve task-specific competence (e.g. language skills, learning objectives), it may lack the depth of contextual understanding and encouragement that human interactions provide. As such, the broader sense of competence that impacts motivation, including social validation and emotional support, may be less effectively provided by GenAI tools. Relatedness has also been reported to have a similar effect size when using GenAI tools, indicating non-significant effects on autonomous motivation. As relatedness is deeply rooted in social interactions, where individuals feel connected and valued by others, building relationships and a sense of belonging typically require human interaction and emotional connections, which GenAI tools cannot fully replicate at the moment. This is supported by Sailer et al. (2017) and Zainuddin (2018), who argued that achieving relatedness requires successful interaction that includes social collaboration between peers and instructors, along with support for engagement amongst these groups.

Overall, our findings confirmed the effectiveness of incorporating GenAI tools to promote autonomous motivation in language learning, mainly through the developmental pathway of autonomy in BPN. The effectiveness of GenAI tools is not solely due to the capabilities of the tools themselves but, more importantly, to the strategic implementation of these tools. In our experiment,

students were granted greater freedom in their English learning with GenAI tools, extending beyond simple language skills to include activities such as developing a learning plan, critically evaluating learning objectives, brainstorming for class activities, and assessing the learning process and outcomes. This approach resulted in students engaging in English learning with genuine interest and consistent commitment, a contrast to a decrease of autonomous motivation seen in traditional teaching. These findings provide valuable insights for language programmes and instructors transitioning to GenAI-assisted language learning or seeking to enhance learners' autonomous motivation. To fully capitalise on its benefits of autonomy, we recommend that instructors implement GenAI tools in a way that integrates students into the decision-making process throughout their learning journey. For example, instructors could provide students with a framework for creating their own personalised learning pathways in collaboration with GenAI tools. A practical implementation might involve asking students to use GenAI tools to draft their own learning objectives at the start of the term and then periodically revising these objectives based on their evolving needs. Instructors could also design "choice boards" for students, allowing them to choose from a variety of activities. For instance, a choice board might include a range of tasks such as vocabulary exercises (e.g., GenAI tools generate flashcards based on recent lessons), grammar tasks (e.g., completing a series of grammar quizzes with explanations from GenAI tools), or writing prompts (e.g., GenAI tools help students brainstorm and outline an essay on a given topic). Each option on the board would be aligned with the learning objectives and provide students with the flexibility to choose activities that best match their individual interests or learning needs. Additionally, students are supposed to be trained to effectively communicate with GenAI tools. This includes teaching them how to reformulate prompts, provide context, and ask follow-up questions to refine their GenAI-assisted learning experience. For example, students could be guided to experiment with different prompt structures, such as "Can you help me explain this concept in simpler terms?" or "Please summarise this article and provide suggestions for further reading?" This approach will help students engage more deeply with the AI chatbot, ensuring that the tool generates relevant, high-quality responses. By successfully interacting with GenAI tools, students are more likely to experience a sense of competence. Lastly, instructors should pay more attention to relatedness in a GenAI-assisted language learning environment. We suggest that instructors design opportunities for collaborative GenAI usage that enhance peer interaction. For example, students could be paired or grouped to complete tasks using GenAI tools, such as generating dialogue scenarios, critiquing language use, or creating group projects that involve collaborative research or writing. These activities would promote interpersonal connections and a sense of shared learning goals. Furthermore, instructors should schedule regular check-ins, not just for academic feedback but also for emotional support, ensuring that students feel recognised and valued in the learning process. One specific strategy might be setting up a "weekly reflection circle." In this setting, students could meet in small groups to share their experiences using GenAI tools during the week. They might discuss challenges they faced, how they overcame them, and what strategies worked best. Instructors could guide these sessions to ensure that students feel supported emotionally, and they could offer feedback on the use of GenAI tools in these reflections. This would not only promote

reflection on academic progress but also reinforce a sense of belonging within the learning community.

6 | Conclusion

Drawing on SDT, this study examines the role of GenAI tools in the English learning motivation of a group of Chinese university students. The results show that integrating this tool into formal language learning can improve students' autonomous motivation over time, as opposed to traditional classroom instruction. This improvement is primarily attributed to the increased autonomy facilitated by the use of GenAI tools. The findings suggest that with careful implementation and evaluation, GenAI tools can be used to enhance language learners' motivation, a necessity in the ever-growing digital learning environment. Our study presents a novel approach to understanding the impact of AI chatbots on language learning motivation, pioneering empirical evidence that the use of GenAI tools is associated with higher levels of autonomous motivation. This finding is significant as it opens new avenues for understanding the broader implications of AI in language learning and suggests potential applications in enhancing learner motivation through technological interventions.

Despite its contributions, this study is subject to some limitations and can be further developed. First, we acknowledge the potential influence of informal GenAI tools use by students in the experimental group. Although the key tenet underlying the research design is not merely the use of GenAI as a tool but rather its implementation into our structured four-step learning cycle during class sessions, the contact of GenAI tool in informal settings remains a confounding variable that could impact the results. Future studies may consider stricter controls or embrace the use of GenAI tools in informal learning when investigating their effect on language learning motivation. Second, the data collected in this study is purely quantitative. Future studies may include qualitative data, such as interviews, to delve into learners' experiences and attitudes toward using GenAI tools in language learning, therefore gaining a more comprehensive understanding of the topic from learners' perceptions. Last but not least, the rapidly evolving nature of digital technology continually transforms the functionalities and usage of tools like GenAI tools. By the time of writing, newly-updated GenAI tools have been launched, featuring multimodal capabilities and improved understanding of human humour and emotional support. These improvements may enhance the effects of competence and relatedness, which earlier versions of GenAI tools failed to provide, and could potentially alter the outcomes of the study. It would be interesting to adopt a flexible and iterative approach to studying motivational dynamics in GenAI-mediated instruction. Researchers could develop longitudinal studies that track how motivational factors evolve as newer GenAI tools with advanced capabilities are introduced into educational settings. For example, studies could investigate how the enhanced emotional intelligence of tools influences students' sense of competence and their emotional engagement with the learning process. The impact of these tools on students' motivation could be measured across multiple dimensions, such as the ability to generate more personalised feedback, detect subtle emotional cues, and maintain engaging, empathetic dialogues. It is also essential to

integrate technological advancements into motivational theory development in real-time. This could involve adjusting existing models of motivation—like SDT—by considering how GenAI can adapt to and respond to diverse learner needs. For instance, with the rise of multimodal tools, future studies might explore how students' use of GenAI tools' multimodal features (text, voice, images, etc.) enhances their sense of autonomy or relatedness in ways that were previously unexamined. Researchers could design new experimental frameworks that assess how these tools support not just cognitive development, but also emotional and social engagement.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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APPENDIX

The basic psychological needs and autonomous motivation adapted from Alamer (2022).

Item
Autonomy
I am able to freely decide my own pace of learning in English
I enjoy exploring different learning strategies and techniques to improve my English skills
I don't feel that I am forced to study English
I feel like I can set my own goals for English proficiency and progress
My English teacher allows my class choose how we approach English learning
Competence
I feel I am capable of learning English
I can be a successful language learner
I am competent enough to meet the challenges and tasks posed in English learning
I feel a sense of accomplishment in my English classes
I feel satisfied with my performance in English learning
Relatedness
I feel a strong sense of belonging to the English learning community
I enjoy participating in group activities and discussions related to English learning
My classmates are willing to help and cooperate with me while learning the language
My English teacher cares about my progress
I seek opportunities to collaborate with others in my English studies
Autonomous motivation
Intrinsic orientation
Because I enjoy learning English
Because of the pleasure I get when English is used
For the satisfaction I feel when I use English
For the enjoyment I experience when I achieve new goal in English learning
Because learning English is a fun activity by itself
Identified orientation
Because learning English is important for my personal growth
Because learning English can open new opportunities and possibilities for me
For the value it holds for my self-development
Because learning English is important for my current and future studies
Because learning English allows me to read and hear English-based materials necessary for my personal success